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Inventor(s)

Srivastava; Abhishek et al.

BACKUP MANAGEMENT OF NON-RELATIONAL DATABASES

Abstract

Methods, systems, and devices for data management are described. For example, techniques for scalable backup solutions for non-relational databases are described. A data management system (DMS) may use different buffer pools for the extraction of data from the non-relational database and the movement (e.g., copying or transferring of data) from the buffers to a remote storage location. A backup process for a collection may involve one or more iterations involving retrieving or selecting a buffer from an empty buffer queue of an agent of the DMS at the non-relational database, filling the buffer with data from the non-relational database to the retrieved buffer, moving the filled buffer to a full buffer queue, moving the data to a remote storage location, and moving the buffer back to the empty buffer queue after moving the data on the buffer to the remote storage location.

Inventors: Srivastava; Abhishek (Bangalore, IN), Kumar; Bittu (Bangalore, IN), Verma; Deepak (Bangalore, IN), Thapliyal; Rahul (Bangalore, IN)

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Background/Summary

FIELD OF TECHNOLOGY

[0001] The present disclosure relates generally to data management, including techniques for backup management of non-relational databases.

BACKGROUND

[0002] A data management system (DMS) may be employed to manage data associated with one or more computing systems. The data may be generated, stored, or otherwise used by the one or more computing systems, examples of which may include servers, databases, virtual machines, cloud computing systems, file systems (e.g., network-attached storage (NAS) systems), or other data storage or processing systems. The DMS may provide data backup, data recovery, data classification, or other types of data management services for data of the one or more computing systems. Improved data management may offer improved performance with respect to reliability, speed, efficiency, scalability, security, or ease-of-use, among other possible aspects of performance.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] FIG. 1 illustrates an example of a computing environment that supports backup management of non-relational databases in accordance with aspects of the present disclosure.

[0004] FIG. 2 shows an example of a non-relational database cluster that supports backup management of non-relational databases in accordance with aspects of the present disclosure.

[0005] FIG. 3 shows an example of a computing environment that supports backup management of non-relational databases in accordance with aspects of the present disclosure.

[0006] FIG. 4 shows an example of a backup process that uses multiple buffer queues that supports backup management of non-relational databases in accordance with aspects of the present disclosure.

[0007] FIG. 5 shows an example of a backup process timeline diagram that supports backup management of non-relational databases in accordance with aspects of the present disclosure.

[0008] FIG. 6 shows an example of a process flow that supports backup management of non-relational databases in accordance with aspects of the present disclosure.

[0009] FIG. 7 shows a block diagram of an apparatus that supports backup management of non-relational databases in accordance with aspects of the present disclosure.

[0010] FIG. 8 shows a block diagram of a data management system that supports backup management of non-relational databases in accordance with aspects of the present disclosure.

[0011] FIG. 9 shows a diagram of a system including a device that supports backup management of non-relational databases in accordance with aspects of the present disclosure.

[0012] FIGS. 10 through 12 show flowcharts illustrating methods that support backup management of non-relational databases in accordance with aspects of the present disclosure.

DETAILED DESCRIPTION

[0013] A data management system (DMS) may include various nodes, clusters, and sub-systems that provide backup and recovery services for customer computing systems or databases. Backup processes may involve capturing snapshots of customer computing systems or databases and storing the snapshots at a storage environment accessible to the DMS. In some cases, the DMS may provide backup and/or recovery services for a non-relational database. For example, a non-relational database may not use a tabular schema of rows and columns and/or may be referred to as a non-SQL or noSQL database. For example, a Mongo database may be a non-relational database. In some examples, a non-relational database may be a document-oriented database that utilizes

JSON-like documents and may include multiple (e.g., thousands of) collections of documents. A non-relational database may be stored at multiple hosts (e.g., a primary host and one or more secondary hosts) which each store a full copy of the data in the database. For example, changes at the primary host may periodically be updated to be reflected at the secondary hosts. Operation logs (oplogs or log snapshots) may capture changes that occur at a given collection at a primary host which may then be replicated to the secondary hosts. Given that a non-relational database may have thousands of collections, per-collection backups may not be scalable for customers of a DMS (e.g., may involve undesirable latencies, among other potential drawbacks).

[0014] Aspects of this disclosure relate to techniques for scalable backup for non-relational databases. To streamline the movement of data from a non-relational database to a remote storage environment as part of the backup process for the non-relational database, the DMS may use different buffer pools for the extraction of data from the non-relational database and the movement (e.g., copying or transferring of data) from the buffers to the remote storage location. For example, a backup process for a collection may involve one or more iterations involving retrieving or selecting a buffer from an empty buffer queue of an agent of the DMS at the non-relational database, filling the buffer with data from the non-relational database to the retrieved buffer, moving the filled buffer to a full buffer queue, moving (e.g., copying) the data to a remote storage location, and moving the buffer back to the empty buffer queue after moving the data on the buffer to the remote storage location. In some examples, the agent may initiate an extractor job or thread (e.g., an extractor job) and a mover job or thread (e.g., a mover job). The extractor job may extract data from the non-relational database and write the data to an empty buffer. The mover job may move data from a buffer in the full buffer queue to the remote storage location. The backup process may be performed iteratively for a collection until all of the data (or the changed or new data for an incremental snapshot) is backed up at the remote storage location. For example, a buffer may be reused once the buffer is moved back to the empty buffer queue. Accordingly, the extraction process and movement process may not block each other. Backup processes may be performed on multiple collections, for example, in parallel, to increase throughput and reduce latency.

[0015] Additionally, or alternatively, the DMS may extract data from the multiple hosts (e.g., the primary host and the one or more secondary hosts or just from two or more secondary hosts) in parallel to reduce the latency associated with a backup of a non-relational database. For example, given 30 collections in a non-relational database to be backed up, the DMS may extract data from 10 of the collections from the primary host, data from 10 other of the collections from a first secondary host, and data from 10 other of the collections from a second secondary host. The DMS may use techniques such as the insertion of special marker documents into the primary host, which will then be replicated into the secondary hosts, to determine whether the secondary hosts are synchronized with the primary host prior to initiating a parallel backup of the Mongo database. Additionally, or alternatively, the DMS may use the oplogs generated by the non-relational database to capture additional changes to a non-relational database after the initiation of the backup of the non-relational database. For example, the additional changes captured by the oplogs may be applied by the DMS on top of a full backup of the non-relational database to generate a consistent snapshot (e.g., all aspects of the snapshot correspond to a single point-in-time and are thus consistent with one another).

[0016] FIG. 1 illustrates an example of a computing environment **100** that supports backup management of non-relational databases in accordance with aspects of the present disclosure. The computing environment **100** may include a computing system **105**, a DMS **110**, and one or more computing devices **115**, which may be in communication with one another via a network **120**. The computing system **105** may generate, store, process, modify, or otherwise use associated data, and the DMS **110** may provide one or more data management services for the computing system **105**. For example, the DMS **110** may provide a data backup service, a data recovery service, a data classification service, a data transfer or replication service, one or more other data management

services, or any combination thereof for data associated with the computing system **105**.

[0017] The network **120** may allow the one or more computing devices **115**, the computing system **105**, and the DMS **110** to communicate (e.g., exchange information) with one another. The network **120** may include aspects of one or more wired networks (e.g., the Internet), one or more wireless networks (e.g., cellular networks), or any combination thereof. The network **120** may include aspects of one or more public networks or private networks, as well as secured or unsecured networks, or any combination thereof. The network **120** also may include any quantity of communications links and any quantity of hubs, bridges, routers, switches, ports or other physical or logical network components.

[0018] A computing device **115** may be used to input information to or receive information from the computing system **105**, the DMS **110**, or both. For example, a user of the computing device **115** may provide user inputs via the computing device **115**, which may result in commands, data, or any combination thereof being communicated via the network **120** to the computing system **105**, the DMS **110**, or both. Additionally, or alternatively, a computing device **115** may output (e.g., display) data or other information received from the computing system **105**, the DMS **110**, or both. A user of a computing device **115** may, for example, use the computing device **115** to interact with one or more user interfaces (e.g., graphical user interfaces (GUIs)) to operate or otherwise interact with the computing system **105**, the DMS **110**, or both. Though one computing device **115** is shown in FIG. **1**, it is to be understood that the computing environment **100** may include any quantity of computing devices **115**.

[0019] A computing device **115** may be a stationary device (e.g., a desktop computer or access point) or a mobile device (e.g., a laptop computer, tablet computer, or cellular phone). In some examples, a computing device **115** may be a commercial computing device, such as a server or collection of servers. And in some examples, a computing device **115** may be a virtual device (e.g., a virtual machine). Though shown as a separate device in the example computing environment of FIG. **1**, it is to be understood that in some cases a computing device **115** may be included in (e.g., may be a component of) the computing system **105** or the DMS **110**.

[0020] The computing system **105** may include one or more servers **125** and may provide (e.g., to the one or more computing devices **115**) local or remote access to applications, databases, or files stored within the computing system **105**. The computing system **105** may further include one or more data storage devices **130**. Though one server **125** and one data storage device **130** are shown in FIG. **1**, it is to be understood that the computing system **105** may include any quantity of servers **125** and any quantity of data storage devices **130**, which may be in communication with one another and collectively perform one or more functions ascribed herein to the server **125** and data storage device **130**.

[0021] A data storage device **130** may include one or more hardware storage devices operable to store data, such as one or more hard disk drives (HDDs), magnetic tape drives, solid-state drives (SSDs), storage area network (SAN) storage devices, or network-attached storage (NAS) devices. In some cases, a data storage device **130** may comprise a tiered data storage infrastructure (or a portion of a tiered data storage infrastructure). A tiered data storage infrastructure may allow for the movement of data across different tiers of the data storage infrastructure between higher-cost, higher-performance storage devices (e.g., SSDs and HDDs) and relatively lower-cost, lower-performance storage devices (e.g., magnetic tape drives). In some examples, a data storage device **130** may be a database (e.g., a relational database), and a server **125** may host (e.g., provide a database management system for) the database.

[0022] A server **125** may allow a client (e.g., a computing device **115**) to download information or files (e.g., executable, text, application, audio, image, or video files) from the computing system **105**, to upload such information or files to the computing system **105**, or to perform a search query related to particular information stored by the computing system **105**. In some examples, a server **125** may act as an application server or a file server. In general, a server **125** may refer to one or

more hardware devices that act as the host in a client-server relationship or a software process that shares a resource with or performs work for one or more clients.

[0023] A server **125** may include a network interface **140**, processor **145**, memory **150**, disk **155**, and computing system manager **160**. The network interface **140** may enable the server **125** to connect to and exchange information via the network **120** (e.g., using one or more network protocols). The network interface **140** may include one or more wireless network interfaces, one or more wired network interfaces, or any combination thereof. The processor **145** may execute computer-readable instructions stored in the memory **150** in order to cause the server **125** to perform functions ascribed herein to the server **125**. The processor **145** may include one or more processing units, such as one or more central processing units (CPUs), one or more graphics processing units (GPUs), or any combination thereof. The memory **150** may comprise one or more types of memory (e.g., random access memory (RAM), static random access memory (SRAM), dynamic random access memory (DRAM), read-only memory (ROM), electrically erasable programmable read-only memory (EEPROM), Flash, etc.). Disk **155** may include one or more HDDs, one or more SSDs, or any combination thereof. Memory **150** and disk **155** may comprise hardware storage devices. The computing system manager **160** may manage the computing system **105** or aspects thereof (e.g., based on instructions stored in the memory **150** and executed by the processor **145**) to perform functions ascribed herein to the computing system **105**. In some examples, the network interface **140**, processor **145**, memory **150**, and disk **155** may be included in a hardware layer of a server **125**, and the computing system manager **160** may be included in a software layer of the server **125**. In some cases, the computing system manager **160** may be distributed across (e.g., implemented by) multiple servers **125** within the computing system **105**.

[0024] In some examples, the computing system **105** or aspects thereof may be implemented within one or more cloud computing environments, which may alternatively be referred to as cloud environments. Cloud computing may refer to Internet-based computing, wherein shared resources, software, and/or information may be provided to one or more computing devices on-demand via the Internet. A cloud environment may be provided by a cloud platform, where the cloud platform may include physical hardware components (e.g., servers) and software components (e.g., operating system) that implement the cloud environment. A cloud environment may implement the computing system **105** or aspects thereof through Software-as-a-Service (SaaS) or Infrastructure-as-a-Service (IaaS) services provided by the cloud environment. SaaS may refer to a software distribution model in which applications are hosted by a service provider and made available to one or more client devices over a network (e.g., to one or more computing devices **115** over the network **120**). IaaS may refer to a service in which physical computing resources are used to instantiate one or more virtual machines, the resources of which are made available to one or more client devices over a network (e.g., to one or more computing devices **115** over the network **120**).

[0025] In some examples, the computing system **105** or aspects thereof may implement or be implemented by one or more virtual machines. The one or more virtual machines may run various applications, such as a database server, an application server, or a web server. For example, a server **125** may be used to host (e.g., create, manage) one or more virtual machines, and the computing system manager **160** may manage a virtualized infrastructure within the computing system **105** and perform management operations associated with the virtualized infrastructure. The computing system manager **160** may manage the provisioning of virtual machines running within the virtualized infrastructure and provide an interface to a computing device **115** interacting with the virtualized infrastructure. For example, the computing system manager **160** may be or include a hypervisor and may perform various virtual machine-related tasks, such as cloning virtual machines, creating new virtual machines, monitoring the state of virtual machines, moving virtual machines between physical hosts for load balancing purposes, and facilitating backups of virtual machines. In some examples, the virtual machines, the hypervisor, or both, may virtualize and make available resources of the disk **155**, the memory, the processor **145**, the network interface

140, the data storage device **130**, or any combination thereof in support of running the various applications. Storage resources (e.g., the disk **155**, the memory **150**, or the data storage device **130**) that are virtualized may be accessed by applications as a virtual disk.

[0026] The DMS **110** may provide one or more data management services for data associated with the computing system **105** and may include DMS manager **190** and any quantity of storage nodes **185**. The DMS manager **190** may manage operation of the DMS **110**, including the storage nodes **185**. Though illustrated as a separate entity within the DMS **110**, the DMS manager **190** may in some cases be implemented (e.g., as a software application) by one or more of the storage nodes **185**. In some examples, the storage nodes **185** may be included in a hardware layer of the DMS **110**, and the DMS manager **190** may be included in a software layer of the DMS **110**. In the example illustrated in FIG. 1, the DMS **110** is separate from the computing system **105** but in communication with the computing system **105** via the network **120**. It is to be understood, however, that in some examples at least some aspects of the DMS **110** may be located within computing system **105**. For example, one or more servers **125**, one or more data storage devices **130**, and at least some aspects of the DMS **110** may be implemented within the same cloud environment or within the same data center.

[0027] Storage nodes **185** of the DMS **110** may include respective network interfaces **165**, processors **170**, memories **175**, and disks **180**. The network interfaces **165** may enable the storage nodes **185** to connect to one another, to the network **120**, or both. A network interface **165** may include one or more wireless network interfaces, one or more wired network interfaces, or any combination thereof. The processor **170** of a storage node **185** may execute computer-readable instructions stored in the memory **175** of the storage node **185** in order to cause the storage node **185** to perform processes described herein as performed by the storage node **185**. A processor **170** may include one or more processing units, such as one or more CPUs, one or more GPUs, or any combination thereof. The memory **150** may comprise one or more types of memory (e.g., RAM, SRAM, DRAM, ROM, EEPROM, Flash, etc.). A disk **180** may include one or more HDDs, one or more SSDs, or any combination thereof. Memories **175** and disks **180** may comprise hardware storage devices. Collectively, the storage nodes **185** may in some cases be referred to as a storage cluster or as a cluster of storage nodes **185**.

[0028] The DMS **110** may provide a backup and recovery service for the computing system **105**. For example, the DMS **110** may manage the extraction and storage of snapshots **135** associated with different point-in-time versions of one or more target computing objects within the computing system **105**. A snapshot **135** of a computing object (e.g., a virtual machine, a database, a filesystem, a virtual disk, a virtual desktop, or other type of computing system or storage system) may be a file (or set of files) that represents a state of the computing object (e.g., the data thereof) as of a particular point in time. A snapshot **135** may also be used to restore (e.g., recover) the corresponding computing object as of the particular point in time corresponding to the snapshot **135**. A computing object of which a snapshot **135** may be generated may be referred to as snappable. Snapshots **135** may be generated at different times (e.g., periodically or on some other scheduled or configured basis) in order to represent the state of the computing system **105** or aspects thereof as of those different times. In some examples, a snapshot **135** may include metadata that defines a state of the computing object as of a particular point in time. For example, a snapshot **135** may include metadata associated with (e.g., that defines a state of) some or all data blocks included in (e.g., stored by or otherwise included in) the computing object. Snapshots **135** (e.g., collectively) may capture changes in the data blocks over time. Snapshots **135** generated for the target computing objects within the computing system **105** may be stored in one or more storage locations (e.g., the disk **155**, memory **150**, the data storage device **130**) of the computing system **105**, in the alternative or in addition to being stored within the DMS **110**, as described below.

[0029] To obtain a snapshot **135** of a target computing object associated with the computing system **105** (e.g., of the entirety of the computing system **105** or some portion thereof, such as one or more

databases, virtual machines, or filesystems within the computing system **105**), the DMS manager **190** may transmit a snapshot request to the computing system manager **160**. In response to the snapshot request, the computing system manager **160** may set the target computing object into a frozen state (e.g., a read-only state). Setting the target computing object into a frozen state may allow a point-in-time snapshot **135** of the target computing object to be stored or transferred.

[0030] In some examples, the computing system **105** may generate the snapshot **135** based on the frozen state of the computing object. For example, the computing system **105** may execute an agent of the DMS **110** (e.g., the agent may be software installed at and executed by one or more servers **125**), and the agent may cause the computing system **105** to generate the snapshot **135** and transfer the snapshot **135** to the DMS **110** in response to the request from the DMS **110**. In some examples, the computing system manager **160** may cause the computing system **105** to transfer, to the DMS **110**, data that represents the frozen state of the target computing object, and the DMS **110** may generate a snapshot **135** of the target computing object based on the corresponding data received from the computing system **105**.

[0031] Once the DMS **110** receives, generates, or otherwise obtains a snapshot **135**, the DMS **110** may store the snapshot **135** at one or more of the storage nodes **185**. The DMS **110** may store a snapshot **135** at multiple storage nodes **185**, for example, for improved reliability. Additionally, or alternatively, snapshots **135** may be stored in some other location connected with the network **120**. For example, the DMS **110** may store more recent snapshots **135** at the storage nodes **185**, and the DMS **110** may transfer less recent snapshots **135** via the network **120** to a cloud environment (which may include or be separate from the computing system **105**) for storage at the cloud environment, a magnetic tape storage device, or another storage system separate from the DMS **110**.

[0032] Updates made to a target computing object that has been set into a frozen state may be written by the computing system **105** to a separate file (e.g., an update file) or other entity within the computing system **105** while the target computing object is in the frozen state. After the snapshot **135** (or associated data) of the target computing object has been transferred to the DMS **110**, the computing system manager **160** may release the target computing object from the frozen state, and any corresponding updates written to the separate file or other entity may be merged into the target computing object.

[0033] In response to a restore command (e.g., from a computing device **115** or the computing system **105**), the DMS **110** may restore a target version (e.g., corresponding to a particular point in time) of a computing object based on a corresponding snapshot **135** of the computing object. In some examples, the corresponding snapshot **135** may be used to restore the target version based on data of the computing object as stored at the computing system **105** (e.g., based on information included in the corresponding snapshot **135** and other information stored at the computing system **105**, the computing object may be restored to its state as of the particular point in time). Additionally, or alternatively, the corresponding snapshot **135** may be used to restore the data of the target version based on data of the computing object as included in one or more backup copies of the computing object (e.g., file-level backup copies or image-level backup copies). Such backup copies of the computing object may be generated in conjunction with or according to a separate schedule than the snapshots **135**. For example, the target version of the computing object may be restored based on the information in a snapshot **135** and based on information included in a backup copy of the target object generated prior to the time corresponding to the target version. Backup copies of the computing object may be stored at the DMS **110** (e.g., in the storage nodes **185**) or in some other location connected with the network **120** (e.g., in a cloud environment, which in some cases may be separate from the computing system **105**).

[0034] In some examples, the DMS **110** may restore the target version of the computing object and transfer the data of the restored computing object to the computing system **105**. And in some examples, the DMS **110** may transfer one or more snapshots **135** to the computing system **105**, and

restoration of the target version of the computing object may occur at the computing system **105** (e.g., as managed by an agent of the DMS **110**, where the agent may be installed and operate at the computing system **105**).

[0035] In response to a mount command (e.g., from a computing device **115** or the computing system **105**), the DMS **110** may instantiate data associated with a point-in-time version of a computing object based on a snapshot **135** corresponding to the computing object (e.g., along with data included in a backup copy of the computing object) and the point-in-time. The DMS **110** may then allow the computing system **105** to read or modify the instantiated data (e.g., without transferring the instantiated data to the computing system). In some examples, the DMS **110** may instantiate (e.g., virtually mount) some or all of the data associated with the point-in-time version of the computing object for access by the computing system **105**, the DMS **110**, or the computing device **115**.

[0036] In some examples, the DMS **110** may store different types of snapshots **135**, including for the same computing object. For example, the DMS **110** may store both base snapshots **135** and incremental snapshots **135**. A base snapshot **135** may represent the entirety of the state of the corresponding computing object as of a point in time corresponding to the base snapshot **135**. An incremental snapshot **135** may represent the changes to the state--which may be referred to as the delta-of the corresponding computing object that have occurred between an earlier or later point in time corresponding to another snapshot **135** (e.g., another base snapshot **135** or incremental snapshot **135**) of the computing object and the incremental snapshot **135**. In some cases, some incremental snapshots **135** may be forward-incremental snapshots **135** and other incremental snapshots **135** may be reverse-incremental snapshots **135**. To generate a full snapshot **135** of a computing object using a forward-incremental snapshot **135**, the information of the forward-incremental snapshot **135** may be combined with (e.g., applied to) the information of an earlier base snapshot **135** of the computing object along with the information of any intervening forward-incremental snapshots **135**, where the earlier base snapshot **135** may include a base snapshot **135** and one or more reverse-incremental or forward-incremental snapshots **135**. To generate a full snapshot **135** of a computing object using a reverse-incremental snapshot **135**, the information of the reverse-incremental snapshot **135** may be combined with (e.g., applied to) the information of a later base snapshot **135** of the computing object along with the information of any intervening reverse-incremental snapshots **135**.

[0037] In some examples, the DMS **110** may provide a data classification service, a malware detection service, a data transfer or replication service, backup verification service, or any combination thereof, among other possible data management services for data associated with the computing system **105**. For example, the DMS **110** may analyze data included in one or more computing objects of the computing system **105**, metadata for one or more computing objects of the computing system **105**, or any combination thereof, and based on such analysis, the DMS **110** may identify locations within the computing system **105** that include data of one or more target data types (e.g., sensitive data, such as data subject to privacy regulations or otherwise of particular interest) and output related information (e.g., for display to a user via a computing device **115**). Additionally, or alternatively, the DMS **110** may detect whether aspects of the computing system **105** have been impacted by malware (e.g., ransomware). Additionally, or alternatively, the DMS **110** may relocate data or create copies of data based on using one or more snapshots **135** to restore the associated computing object within its original location or at a new location (e.g., a new location within a different computing system **105**). Additionally, or alternatively, the DMS **110** may analyze backup data to ensure that the underlying data (e.g., user data or metadata) has not been corrupted. The DMS **110** may perform such data classification, malware detection, data transfer or replication, or backup verification, for example, based on data included in snapshots **135** or backup copies of the computing system **105**, rather than live contents of the computing system **105**, which may beneficially avoid adversely affecting (e.g., infecting, loading, etc.) the computing system **105**.

[0038] In some examples, the DMS **110**, and in particular the DMS manager **190**, may be referred to as a control plane. The control plane may manage tasks, such as storing data management data or performing restorations, among other possible examples. The control plane may be common to multiple customers or tenants of the DMS **110**. For example, the computing system **105** may be associated with a first customer or tenant of the DMS **110**, and the DMS **110** may similarly provide data management services for one or more other computing systems associated with one or more additional customers or tenants. In some examples, the control plane may be configured to manage the transfer of data management data (e.g., snapshots **135** associated with the computing system **105**) to a cloud environment **195** (e.g., Microsoft Azure or Amazon Web Services). In addition, or as an alternative, to being configured to manage the transfer of data management data to the cloud environment **195**, the control plane may be configured to transfer metadata for the data management data to the cloud environment **195**. The metadata may be configured to facilitate storage of the stored data management data, the management of the stored management data, the processing of the stored management data, the restoration of the stored data management data, and the like.

[0039] Each customer or tenant of the DMS **110** may have a private data plane, where a data plane may include a location at which customer or tenant data is stored. For example, each private data plane for each customer or tenant may include a node cluster **196** across which data (e.g., data management data, metadata for data management data, etc.) for a customer or tenant is stored. Each node cluster **196** may include a node controller **197** which manages the nodes **198** of the node cluster **196**. As an example, a node cluster **196** for one tenant or customer may be hosted on Microsoft Azure, and another node cluster **196** may be hosted on Amazon Web Services. In another example, multiple separate node clusters **196** for multiple different customers or tenants may be hosted on Microsoft Azure. Separating each customer or tenant's data into separate node clusters **196** provides fault isolation for the different customers or tenants and provides security by limiting access to data for each customer or tenant.

[0040] The control plane (e.g., the DMS **110**, and specifically the DMS manager **190**) manages tasks, such as storing backups or snapshots **135** or performing restorations, across the multiple node clusters **196**. For example, as described herein, a node cluster **196-a** may be associated with the first customer or tenant associated with the computing system **105**. The DMS **110** may obtain (e.g., generate or receive) and transfer the snapshots **135** associated with the computing system **105** to the node cluster **196-a** in accordance with a service level agreement for the first customer or tenant associated with the computing system **105**. For example, a service level agreement may define backup and recovery parameters for a customer or tenant such as snapshot generation frequency, which computing objects to backup, where to store the snapshots **135** (e.g., which private data plane), and how long to retain snapshots **135**. As described herein, the control plane may provide data management services for another computing system associated with another customer or tenant. For example, the control plane may generate and transfer snapshots **135** for another computing system associated with another customer or tenant to the node cluster **196-n** in accordance with the service level agreement for the other customer or tenant.

[0041] To manage tasks, such as storing backups or snapshots **135** or performing restorations, across the multiple node clusters **196**, the control plane (e.g., the DMS manager **190**) may communicate with the node controllers **197** for the various node clusters via the network **120**. For example, the control plane may exchange communications for backup and recovery tasks with the node controllers **197** in the form of transmission control protocol (TCP) packets via the network **120**.

[0042] The DMS **110** may provide backup and recovery services for a non-relational database (e.g., the computing system **105**) may be a non-relational database and the DMS **110** may capture snapshots **135** of the non-relational database. The non-relational database may be stored at multiple hosts (e.g., a primary host and one or more secondary hosts) which each store a full copy of the

data in the database. For example, different hosts may be different servers, different virtual machines, or different storage nodes. Data in the non-relational database may be organized as collections of documents (e.g., JSON-like documents). As the non-relational database may have thousands of collections, the DMS **110** may implement techniques for scalable backup of the non-relational database.

[0043] For example, to streamline the movement of data from the non-relational database to a remote storage environment (e.g., one or more storage nodes **185** at the DMS **110** or one or more node clusters **196** at the cloud environment **195**) as part of the backup process for the non-relational database, the DMS **110** may use different buffer pools for the extraction of data from the non-relational database and the movement (e.g., copying or transferring of data) from the buffers to the remote storage location. For example, a backup process for a collection may involve one or more iterations involving retrieving or selecting a buffer from an empty buffer queue of an agent of the DMS **110** at the non-relational database, filling the buffer with data from the non-relational database to the retrieved buffer, moving the filled buffer to a full buffer queue, moving (e.g., copying) the data to a remote storage location, and moving the buffer back to the empty buffer queue after moving the data on the buffer to the remote storage location. In some examples, the agent may initiate an extractor job and a mover job. The extractor job may extract data from the non-relational database and write the data to an empty buffer. The mover job may move data from a buffer in the full buffer queue to the remote storage location. The backup process may be performed iteratively for a collection until all of the data (or the modified or new data for an incremental snapshot) is backed up at the remote storage location. For example, a buffer may be reused once the buffer is moved back to the empty buffer queue. Accordingly, the extraction process and movement process may not block each other. Backup processes may be performed on multiple collections, for example, in parallel, to increase throughput and reduce latency.

[0044] Additionally, or alternatively, the DMS **110** may extract data from the multiple hosts (e.g., the primary host and the one or more secondary hosts) in parallel to reduce the latency associated with a backup of a non-relational database. For example, given 30 collections in a non-relational database to be backed up, the DMS **110** may extract data from 10 of the collections from the primary host, data from 10 other of the collections from a first secondary host, and data from 10 other of the collections from a second secondary host. The DMS **110** may use techniques such as the insertion of special marker documents into the primary host, which will then be replicated into the secondary hosts, to determine whether the secondary hosts are synchronized with the primary host prior to initiating a parallel backup of the Mongo database. Additionally, or alternatively, the DMS **110** may use the oplogs generated by the non-relational database to capture additional changes to a non-relational database after the initiation of the backup of the non-relational database.

[0045] FIG. **2** shows an example of a diagram **200** of a non-relational database cluster **205** that supports backup management of non-relational databases in accordance with aspects of the present disclosure. The diagram **200** may implement or may be implemented by one or more aspects of the computing environment **100**. For example, a DMS **110** may backup (e.g., may capture snapshots of) data stored at the non-relational database cluster **205**.

[0046] The non-relational database cluster **205** may include multiple non-relational database hosts **210** that include copies of the same data (e.g., include synchronized copies of a non-relational database). For example, the non-relational database cluster **205** may include a non-relational database host A **210-a** and a non-relational database host B **210-b**. The non-relational database host A **210-a** may store a first copy of the non-relational database and the non-relational database host B **210-b** may store a second copy of the non-relational database. To maintain copies of the same data (e.g., the non-relational database), data may be replicated from the non-relational database host A **210-a** to the non-relational database host B **210-b** (e.g., the non-relational database host A **210-a** may be a primary host and the non-relational database host B **210-b** may be a secondary host). In

some examples, the non-relational database host A **210-a** and the non-relational database host B **210-b** may host a Mongo database.

[0047] As hosting a non-relational database, the non-relational database host A **210-a** and non-relational database host B **210-b** may store collections of data including one or more documents. For example, the non-relational database host A **210-a** may store a collection A which includes documents A-1 through A-n, a collection B which includes documents B-1 through B-n, and a collection N which includes documents N-1 through N-n. The non-relational database host B **210-b** may store a copy of the data stored at the non-relational database host A **210-a**, and accordingly the non-relational database host B **210-b** may store a collection A' which includes documents A-1 through A-n (e.g., the collection A' may be a copy of the collection A), a collection B' which includes documents B-1 through B-n (e.g., the collection B' may be a copy of the collection B), and a collection N' which includes documents N-1 through N-n (e.g., the collection N' may be a copy of the collection N). In some examples, a document in a non-relational database may be a key value pair list or array or a nested document. In some examples, a non-relational database may store data records as binary JSON (BSON) documents (e.g., a BSON may be a binary representation of a JSON document).

[0048] FIG. 3 shows an example of a computing environment **300** that supports backup management of non-relational databases in accordance with aspects of the present disclosure. The computing environment **300** may implement or may be implemented by one or more aspects of the computing environment **100** or the diagram **200**. For example, the computing environment **300** includes a DMS **110-a** which may be an example of the DMS **110** as described herein. The computing environment **300** may include a non-relational database cluster **205-a** which may be an example of a non-relational database cluster **205** as described with reference to FIG. 2. The computing environment **300** may include a storage environment **305**, which may be hosted locally at the DMS **110-a** (e.g., one or more storage nodes **185** at the DMS **110**) or may be accessible to the DMS **110-a** (e.g., or one or more node clusters **196** at the cloud environment **195**).

[0049] The database cluster **205-a** may include non-relational database hosts **310**. The non-relational database hosts **310** may be examples of the non-relational database hosts **210** as described with reference to FIG. 2. For example, the non-relational database host **310-a** may be a primary host for a non-relational database, and the non-relational database host **310-b** and the non-relational database host **310-c** may be secondary hosts for the non-relational database. For example, data stored in the non-relational database host **310-a** may be replicated to the non-relational database host **310-b** and the non-relational database host **310-c**. For example, each of the non-relational database host **310-a**, the non-relational database host **310-b**, and the non-relational database host **310-c** may store copies of nine collections as shown in FIG. 3.

[0050] The DMS **110-a** may provide backup and recovery services for the database cluster **205-a**. For example, the DMS **110-a** may capture snapshots of the non-relational database stored at the non-relational database hosts **310** (e.g., the collections 1 through 9). For example, the DMS **110-a** may communicate with an agent **315** of the DMS **110-a** at each of the non-relational database host **310-b** and the non-relational database host **310-c**. For example, the DMS **110-a** may instantiate or communicate with an agent **315-a** at the non-relational database host **310-a** via a connection **325-a** (e.g., via a network **120** as described with reference to FIG. 1) with the non-relational database host **310-a**. The DMS **110-a** may instantiate or communicate with an agent **315-b** at the non-relational database host **310-b** via a connection **325-b** (e.g., via a network **120** as described with reference to FIG. 1) with the non-relational database host **310-b**. The DMS **110-a** may instantiate or communicate with an agent **315-c** at the non-relational database host **310-c** via a connection **325-c** (e.g., via a network **120** as described with reference to FIG. 1) with the non-relational database host **310-c**.

[0051] In some examples, to decrease latency associated with backing up the data at the non-relational database hosts **310**, the DMS **110-a** may extract data from the non-relational database

hosts **310** in parallel. For example, the DMS **110-a** may extract data from collections 1 through 3 from the non-relational database host **310-a**, the DMS **110-a** may extract data from collections 4 through 6 from the non-relational database host **310-b**, and the DMS **110-a** may extract data from collections 7 through 9 from the non-relational database host **310-c**. Parallel extraction from multiple hosts may enable the DMS **110-a** to leverage resources of the multiple hosts and speed up the backup process (e.g., using an agent **315** at each host). Although shown as three non-relational database hosts **310**, data stored on a non-relational database host **310** may be replicated to any quantity of non-relational database hosts **310** (e.g., 2, 3, 4, 5, etc.), and a DMS **110-a** may correspondingly extract data from the quantity of non-relational database hosts in parallel.

[0052] In some examples, prior to initiating a backup process from multiple hosts in parallel, the DMS **110-a** may check whether the hosts are synchronized. For example, the DMS **110-a** may check whether the data at the non-relational database host **310-a** has been replicated to the non-relational database host **310-b** and the non-relational database host **310-c**. For example, the DMS **110-a** (or the agent **315-a** of the DMS **110-a**) may insert a marker document that includes a timestamp into a known location (e.g., one of the collections) at the non-relational database host **310-a**. For example, the known location may be a collection managed by the DMS **110-a**. Once the DMS **110-a** identifies that the marker document including the timestamp is at the same location at the non-relational database host **310-b**, the DMS **110-a** may determine that the non-relational database host **310-b** is synchronized to the non-relational database host **310-a** as of the timestamp of the marker document and that the insertion of the marker document is included in a log snapshot for the non-relational database host **310-a**. For example, to perform a replication after an initial replication, the primary host (e.g., the non-relational database host **310-a**) may capture a log snapshot indicating changes to documents (e.g., new documents, deleted documents, or modified documents) in the collections of the non-relational database host **310-a** since a prior log snapshot. The changes indicated by the log snapshot may then be replicated to the secondary hosts (e.g., the non-relational database host **310-b** and the non-relational database host **310-c**) to maintain synchronization between the primary and secondary hosts. Thus, the insertion of a marker document into the primary host may be identified in a log snapshot.

[0053] Upon determining that the non-relational database host **310-b** is synchronized to the non-relational database host **310-a** as of the timestamp of the marker document, the DMS **110-a** may initiate a parallel backup process from the non-relational database host **310-a** and the non-relational database host **310-b**. Similarly, once the DMS **110-a** identifies that the marker document including the timestamp is at the same location at the non-relational database host **310-c**, the DMS **110-a** may determine that the non-relational database host **310-b** is synchronized to the non-relational database host **310-a** as of the timestamp of the marker document.

[0054] For a parallel extraction, the collections (e.g., collections 1 through 9) may be divided into collections groups. For example, a collection group 1 that may be extracted from the non-relational database host **310-a** may include collections 1-3, a collection group 2 that may be extracted from the non-relational database host **310-b** may include collections 4-6, and a collection group 3 that may be extracted from the non-relational database host **310-c** may include collections 7-9. In some examples, the collections may be divided into collections groups based on collection indices (e.g., such that the DMS **110-a** will extract an equal or approximately equal quantity of collections from each non-relational database host). In some examples, the collections may be divided into collections groups based on data size (e.g., some collections may be larger than other collections, and thus the DMS **110-a** may extract more collections from one non-relational database host **310** than another but the DMS **110-a** may extract an approximately equal amount of data from each non-relational database). Each collection group may have a separate extraction path (e.g., the connection **325-a** for the collection group from the non-relational database host **310-a**, the connection **325-b** for the collection group from the non-relational database host **310-b**, and the connection **325-c** for the collection group from the non-relational database host **310-c**). For

example, the separate extraction paths may be separate network file system (NFS) export paths. For example, the data may be extracted and sent to the storage environment **305** in an AF2 file format.

[0055] In some examples, the DMS **110-a** may include a discovery job **330** which may identify the non-relational databases which host the replicated data. In some examples, the discovery job **330** may identify that the non-relational database hosts **310** are synchronized.

[0056] As described herein, the non-relational database host **310-a** may capture log snapshots (e.g., periodically) to identify changes to the non-relational database host **310-a** and replicate the changes to the non-relational database host **310-b** and the non-relational database host **310-c**. In some examples, when the DMS **110-a** obtains an indication to capture a full snapshot of the non-relational database stored at the non-relational database hosts **310**, the DMS **110-a** may trigger (e.g., the agent **315** may trigger) an on-demand log snapshot for the non-relational database. For example, the DMS **110-a** may obtain an indication to trigger a full snapshot of the non-relational database stored at the non-relational database hosts **310** based on a service level agreement (SLA) with a customer of the DMS **110-a** associated with the non-relational database, where the SLA schedules periodic backups of the non-relational database stored at the non-relational database hosts **310**. As another example, the DMS **110-a** may receive an indication to capture a snapshot of the non-relational database stored at the non-relational database hosts **310** from an administrative account associated with a customer of the DMS **110-a** associated with the non-relational database (e.g., from a computing device **115** associated with the administrative account).

[0057] In some examples, the DMS **110-a** may wait (e.g., the agents **315** may wait) until the on-demand log snapshot completes to begin the full snapshot (e.g., parallel extraction from the non-relational database hosts **310**). In some examples, waiting for a log snapshot to complete may guarantee that any change events happening to the data source for the non-relational database stored at the non-relational database hosts **310** are captured in the snapshot of the non-relational database. After the on-demand log snapshot completes, the DMS **110-a** may check whether the DMS **110-a** may start a full extraction from the non-relational database hosts **310**. For example, the DMS **110-a** may perform an application programming interface (API) call to a given host that may check (e.g., verify that) the host is capturing change events related to the corresponding non-relational database. For example, the API call may be a “canStartMongodbFullbackupExtraction” API call. In some examples, the API may indicate a timestamp of a last marker document inserted into the primary host, and the API may return an indication of whether a secondary host includes the marker document with the indicated timestamp. If the API call indicates that the secondary host includes the marker document with the indicated timestamp, the DMS **110-a** may determine that the secondary host is synchronized with the primary host and data may be extracted in parallel from the primary host and the secondary host.

[0058] In some examples, in response to verifying that the host is capturing change events related to the corresponding non-relational database hosts **310**, the DMS **110-a** may begin the full extraction process from that host. For example, the DMS **110-a** may cause the agent **315-a** to start the full extraction process of the collection group from the non-relational database host **310-a** in response to verifying that the host of the non-relational database host **310-a** is capturing change events related to the non-relational database host **310-a**. In some examples, to extract data from the collection group from the non-relational database host **310-a**, the agent **315-a** may: export the NFS path for the collection group to the DMS **110-a** and mount the directory on the host of the non-relational database host **310-a**. In response, the DMS **110-a** may send an API call (e.g., “startMongodbFullbackupExtraction”) to the host of the non-relational database host **310-a** (e.g., to the agent **315-a**) that initiates the extraction process for the collection group. The API call may contain the list of collections to be extracted from the non-relational database host **310-a** (e.g., the collection group). The DMS **110-a** may perform a similar process to verify and extract data from the other non-relational database hosts **310** (e.g., the non-relational database host **310-b** and the non-relational database host **310-c**). In some examples, the DMS **110-a** may perform parallel

extractions of collections from a same non-relational database host **310** based on capabilities of the non-relational database host **310**. For example, the DMS **110-a** may extract the collection 1 and the collection 2 from the non-relational database host **310-a** in parallel.

[0059] In some examples, the DMS **110-a** may wait for the extraction process to complete. The DMS **110-a** may perform an API call to check the progress of the extraction process for a given non-relational database host **310** (e.g., an API call “getFullbackupExtractionStatus.”) Once the full extraction process is complete for a given non-relational database (e.g., for the collection group from a given non-relational database), the DMS **110-a** may wait for a log snapshot to be completed for the non-relational database host **310** after the completion of the full extraction. The log snapshot to be completed for the non-relational database host **310** after the completion of the full extraction may track any changes that occurred to the non-relational database host **310** (e.g., at the source of data for the non-relational database host **310**) during the full extraction process. Once the log snapshot is completed, the DMS **110-a** may perform post processing on the data in the full extraction based on the log snapshot to update the data in the full extraction based on the changes that occurred during the full extraction process as reflected by the log snapshot. The updated data in the full extraction may be exposed as a full snapshot of the non-relational database stored at the non-relational database host **310** (e.g., for the collection group) and may be stored in the storage environment **305**.

[0060] FIG. 4 shows an example of a diagram of a backup process **400** that uses multiple buffer queues that supports backup management of non-relational databases in accordance with aspects of the present disclosure. The diagram of the backup process **400** may implement or may be implemented by one or more aspects of the computing environment **100**, the diagram **200**, or the computing environment **300**. For example, the diagram of the backup process **400** may include a storage environment **405**, which may be an example of a storage environment **305** as described with reference to FIG. 3.

[0061] A DMS **110** may provide backup and recovery services for a non-relational database **410**. For example, an agent **315** of the DMS **110-a** at the host of the non-relational database **410** may extract data from one or more collections at the non-relational database **410** and move the data (e.g., copy the data) to the storage environment **405** (e.g., a remote storage environment). In some examples, the agent **315** may initiate an extractor **415** and a mover **430**. The extractor **415** and the mover **430** may operate based on a producer/consumer paradigm, where the extractor **415** is the producer and the mover **430** is the consumer. The extractor **415** may extract data from the non-relational database **410** and may write data to a buffer of the agent (e.g., at the host of the non-relational database **410**). The extractor **415** may be a single threaded execution. The mover **430** may move the data from the buffer to the storage environment **405**. The mover **430** may be a single threaded execution. To reduce latency (e.g., to prevent blocking each other), the extractor **415** and the mover **430** may use separate buffer pools (e.g., buffer queues). Accordingly, the agent **315** may use the extractor **415** and the mover **430** to perform one or more iterations of a backup process for a given collection to extract and move data from the collection of the non-relational database **410** to the storage environment until all of the information in the collection is moved to the storage environment **405** (or for an incremental snapshot, all of the new or modified data in the collection is moved to the storage environment **405**). The extractor **415** may use a cursor of the driver of the host of the non-relational database **410** to iterate through the documents in the relevant collection and read and extract the documents.

[0062] In some examples, when the agent **315** is indicated by the DMS **110** to initiate a backup for a collection at the non-relational database **410**, the agent may initiate the extractor **415** and the mover **430**. The agent **315** may perform separate backup processes for each collection at a non-relational database, and accordingly, the agent **315** may initiate separate extractors and movers for each collection to be backed up.

[0063] At a first step of an iteration (S1), the extractor may retrieve (e.g., consume) a buffer from

an empty buffer queue **420** of the agent **315**. A buffer may be a pre-allocated memory of the agent **315** (e.g., 32 MB). The size of the buffer may be configurable. At a second step (s2) of the iteration the extractor **415** may write data from the collection of the non-relational database **410** to the retrieved buffer (e.g., until the buffer is full). At a third step (s3) of the iteration, the extractor **415** may move the buffer to a full buffer queue **425** of the agent **315**. At a fourth step (s4) of the iteration, the mover **430** may consume (e.g., retrieve or select a buffer from the full buffer queue **425**) and may move (e.g., copy) data from the consumed buffer to the storage environment **405**. For example, the mover **430** may move the data using an NFS as described herein. At a fifth step (s5) of the iteration, after the data on the buffer is moved to the storage environment **405** (e.g., published to the storage environment **405**), the mover **430** may move the buffer back to the empty buffer queue so that the buffer may be reused by the extractor **415**. The extractor **415** and the mover **430** may perform the iterations until all the data from the corresponding collection is moved to the storage environment **405** (or for an incremental snapshot, all of the new or modified data in the collection is moved to the storage environment **405**).

[0064] FIG. 5 shows an example of a backup process timeline diagram **500** that supports backup management of non-relational databases in accordance with aspects of the present disclosure. The backup process timeline diagram **500** may implement or may be implemented by one or more aspects of the computing environment **100**, the diagram **200**, the computing environment **300**, or the diagram of the backup process **400**. For example, the backup process timeline diagram **500** may include a non-relational database **510**, which may be an example a non-relational database **410** as described with reference to FIG. 4. The backup process timeline diagram **500** may include a DMS **110-b**, which may be an example of a DMS **110** as described herein.

[0065] As described herein, a DMS **110-b** may cause an agent of the DMS **110-b** at the non-relational database **510** to perform a full extraction **515** of data at the non-relational database **510** (e.g., from one or more collections of the non-relational database **510** at a host of the non-relational database **510**). The full extraction **515** may be transferred to the DMS **110-b** (e.g., to a storage environment at the DMS **110-b** or accessible to the DMS **110-b**).

[0066] As described herein, the non-relational database **510** may capture log snapshots **505** (e.g., periodically) to identify changes to the non-relational database **510** and replicate the changes to other hosts of the non-relational database. For example, log snapshots **505** may be captured at a high frequency, such as every 15 minutes or every 30 minutes. Log snapshots **505** may capture change events that occur at the non-relational database **510**.

[0067] As shown in the timeline **530**, the full extraction may begin at t15 once the log snapshots have been running. A first log snapshot **505-a** may be initiated at t0 prior to initiation of the full extraction at t15. A second log snapshot **505-b** may be initiated at t5 prior to initiation of the full extraction at t15. A third log snapshot **505-c** may be initiated at t20 during the full extraction at t15. A fourth log snapshot **505-d** may be initiated at t35 during the full extraction at t15.

[0068] The full extraction **515** may be combined with one or more log snapshots by a post processor **520** (e.g., at the DMS **110**) to create a snapshot **525** of the non-relational database **510** (e.g., of the collections corresponding to the full extraction **515**). For example, as shown in FIG. 5, the log snapshots **505** begin before the full extraction **515** is initiated at t15. As shown in FIG. 5, the full extraction **515** is completed at t50. Accordingly, the full extraction **515** may not include changes that occur to the non-relational database **510** during the full extraction **515** (e.g., between t15 and t50). The post processor may generate the snapshot **525** based on updating the full extraction **515** to include the changes that occurred between t15 and t50 to the non-relational database **510** as reflected by the log snapshots **505** (e.g., the second log snapshot **505-b**, the third log snapshot **505-c**, and the fourth log snapshot **505-d**).

[0069] FIG. 6 shows an example of a process flow **600** that supports backup management of non-relational databases in accordance with aspects of the present disclosure. The process flow **600** may implement or may be implemented by one or more aspects of the computing environment **100**, the

diagram **200**, the computing environment **300**, the diagram of the backup process **400**, or the backup process timeline diagram **500**. For example, the process flow **600** may include a DMS **110-c**, which may be an example of a DMS **110** as described herein. The process flow **600** may include a remote storage environment **605**, which may be an example of a storage environment **305** as described with reference to FIG. 3 or a storage environment **405** as described with reference to FIG. 4. The process flow **600** may include a host **610** of a non-relational database, which may be an example of a non-relational database host **210** as described with reference to FIG. 2 or a non-relational database host **310** as described with reference to FIG. 3. In the following description of the process flow **600**, operations between the DMS **110-b**, the remote storage environment **605**, and the host **610** may be added, omitted, or performed in a different order (with respect to the exemplary order shown).

[0070] The DMS **110-c** may be configured to manage backup operations for a non-relational database. The non-relational database may be hosted at the host **610**. At **615**, the DMS **110-c** may obtain an indication to capture a snapshot of the non-relational database from the host **610** of the non-relational database.

[0071] At **620**, the DMS **110-c** may cause an agent of the DMS **110-c** at the host **610** to perform one or more iterations of a backup process until the snapshot is captured.

[0072] At **625**, the agent of the DMS **110-c** may perform the one or more iterations of the backup process until the snapshot is captured.

[0073] One iteration of the backup process may include: at **630**, retrieving, based on a presence of uncaptured data in the non-relational database, a buffer at the agent from an empty buffer queue for the agent; at **635** writing, based on the buffer being retrieved, data from the non-relational database into the buffer until the buffer is full, where the data includes at least a portion of the uncaptured data; at **640** transitioning, based upon filling of the buffer with the data, the buffer to a full buffer queue for the agent; at **645** moving, based on the buffer being transitioned to the full buffer queue, the data from the buffer to the remote storage environment **605** accessible to the DMS **110-c**; and at **650** transitioning, based on completion of moving the data to the remote storage environment **605**, the buffer from the full buffer queue to the empty buffer queue.

[0074] In some examples, the snapshot is associated with a set of collections of data stored at both the host **610** and a second host of the non-relational database. The DMS **110-c** may cause performance of one or more second iterations of the backup process by a second agent of the DMS **110-c** at the second host until a second portion of the snapshot is captured from the second host, where performance by the agent of the DMS of the one or more iterations of the backup process until the snapshot is captured involves performance of the one or more iterations until a first portion of the snapshot is captured from the host **610**. The first portion may include a first subset of collections of the set of collections, and the second portion may include a second subset of collections of the set of collections. In some examples, the one or more iterations and the one or more second iterations may be performed in parallel. In some examples, the set of collections of data is also stored at a third host of the non-relational database, and the DMS **110-c** may cause performance of one or more third iterations of the backup process by a third agent of the DMS **110-c** at the third host until a third portion of the snapshot is captured from the second host, where the third portion includes a third subset of collections of the set of collections. In some examples, the host **610** is a primary host of the non-relational database and the second host and the third host are secondary hosts of the non-relational database that are configured to store respective copies of the non-relational database from the host.

[0075] In some examples, the DMS **110-c** may determine a presence of a synchronization indicator document in a collection at the host, and causing performance of the one or more iterations is based on determination of the presence of the synchronization indicator document. In some examples, the DMS **110-c** may cause insertion, prior to the determination of the presence of the synchronization indicator document in the collection at the host, of the synchronization indicator document into the

collection at a second host of the non-relational database, where the second host and the host both store respective copies of the non-relational database. In some examples, the DMS **110-c** may determine whether a timestamp indicated by the synchronization indicator document is later than a threshold time, and causing performance of the one or more iterations is based on a determination that the timestamp is later than the threshold time.

[0076] In some examples, the DMS **110-c** may cause initiation, based on the indication to capture the snapshot, of an extractor job and a mover job for a first collection from a set of collections of data associated with the snapshot. The extractor job may perform for the first collection and as part of the one or more iterations of the backup process: the retrieving the buffer from the empty buffer queue; the writing the data from the non-relational database into the buffer; and the transitioning the buffer to the full buffer queue. The mover job may perform, for the first collection and as part of the one or more iterations of the backup process: the moving the data from the buffer to the remote storage environment; and the transitioning the buffer from the full buffer queue to the empty buffer queue. In some examples, the DMS **110-c** may cause initiation, based on the indication to capture the snapshot, of a second extractor job and a second mover job for a second collection from the set of collections, and the DMS **110-c** may cause performance by the agent of the DMS one or more second iterations of the backup process. An iteration of the one or more second iterations may involve: retrieving, by the second extractor job and based on a presence of second uncaptured data in the second collection, a second buffer at the agent from a second empty buffer queue for the agent; writing, by the second extractor job and based on the second buffer being retrieved, second data from the second collection into the second buffer until the second buffer is full, where the second data includes at least a second portion of the second uncaptured data; transitioning, by the second extractor job and based upon filling of the second buffer with the second data, the second buffer to a second full buffer queue for the agent; moving, by the second mover job and based on the second buffer being transitioned to the second full buffer queue, the second data from the second buffer to the remote storage environment; and transitioning, by the second mover job and based on completion of moving the second data to the remote storage environment, the second buffer from the second full buffer queue to the second empty buffer queue. In some examples, selecting the buffer from the empty buffer queue by the extractor job is based on the presence of the uncaptured data in the first collection. In some examples, the one or more iterations of the backup process for the first collection and the one or more second iterations of the backup process for the second collection are performed in parallel.

[0077] In some examples, the DMS **110-c** may update the snapshot based on one or more change log files (e.g., log snapshots) that indicate one or more changes to the non-relational database during a time period in which the one or more iterations are performed. In some examples, the DMS **110-c** may perform an API call to the host, and the DMS may receive, from the host **610** in response to the API call, an indication that the host **610** is configured to capture change log files for the non-relational database. Causing performance of the one or more iterations may be based on reception of the indication that the host **610** is configured to capture the change log files.

[0078] In some examples, writing data from the non-relational database into the buffer at **635** involves writing one or more documents into the buffer.

[0079] In some examples, obtaining the indication to capture the snapshot at **615** is based on determining that a time period since a last snapshot of the non-relational database satisfies a threshold time period (e.g., the DMS **110-c** may be scheduled to capture periodic snapshots of the non-relational database).

[0080] In some examples, obtaining the indication to capture the snapshot at **615** is based on receiving an indication to capture the snapshot from a computing device (e.g., a computing device **115**) associated with an administrative account associated with the non-relational database.

[0081] FIG. 7 shows a block diagram **700** of a system **705** that supports backup management of non-relational databases in accordance with aspects of the present disclosure. In some examples,

the system **705** may be an example of aspects of one or more components described with reference to FIG. **1**, such as a DMS **110**. The system **705** may include an input interface **710**, an output interface **715**, and a DMS manager **720**. The system **705** may also include one or more processors. Each of these components may be in communication with one another (e.g., via one or more buses, communications links, communications interfaces, or any combination thereof).

[0082] The input interface **710** may manage input signaling for the system **705**. For example, the input interface **710** may receive input signaling (e.g., messages, packets, data, instructions, commands, or any other form of encoded information) from other systems or devices. The input interface **710** may send signaling corresponding to (e.g., representative of or otherwise based on) such input signaling to other components of the system **705** for processing. For example, the input interface **710** may transmit such corresponding signaling to the DMS manager **720** to support backup management of non-relational databases. In some cases, the input interface **710** may be a component of a network interface **925** as described with reference to FIG. **9**.

[0083] The output interface **715** may manage output signaling for the system **705**. For example, the output interface **715** may receive signaling from other components of the system **705**, such as the DMS manager **720**, and may transmit such output signaling corresponding to (e.g., representative of or otherwise based on) such signaling to other systems or devices. In some cases, the output interface **715** may be a component of a network interface **925** as described with reference to FIG. **9**.

[0084] For example, the DMS manager **720** may include a snapshot scheduling manager **725**, a backup process iteration manager **730**, or any combination thereof. In some examples, the DMS manager **720**, or various components thereof, may be configured to perform various operations (e.g., receiving, monitoring, transmitting) using or otherwise in cooperation with the input interface **710**, the output interface **715**, or both. For example, the DMS manager **720** may receive information from the input interface **710**, send information to the output interface **715**, or be integrated in combination with the input interface **710**, the output interface **715**, or both to receive information, transmit information, or perform various other operations as described herein.

[0085] The DMS manager **720** may support data management in accordance with examples as disclosed herein. The snapshot scheduling manager **725** may be configured as or otherwise support a means for obtaining, by a DMS configured to manage backup operations for a non-relational database, an indication to capture a snapshot of the non-relational database from a host of the non-relational database. The backup process iteration manager **730** may be configured as or otherwise support a means for causing, by the DMS, performance by an agent of the DMS at the host of one or more iterations of a backup process until the snapshot is captured. In some examples, to perform an iteration of the backup process, the empty buffer retrieval manager **735** may be configured as or otherwise support a means for retrieving, based on a presence of uncaptured data in the non-relational database, a buffer at the agent from an empty buffer queue for the agent, the buffer writing manager **740** may be configured as or otherwise support a means for writing, based on the buffer being retrieved, data from the non-relational database into the buffer until the buffer is full, where the data includes at least a portion of the uncaptured data, the full buffer transition manager **745** may be configured as or otherwise support a means for transitioning, based at least in part upon filling of the buffer with the data, the buffer to a full buffer queue for the agent, the data mover manager **750** may be configured as or otherwise support a means for moving, based on the buffer being transitioned to the full buffer queue, the data from the buffer to a remote storage environment accessible to the DMS, and the empty buffer transition manager **755** may be configured as or otherwise support a means for transitioning, based on completion of moving the data to the remote storage environment, the buffer from the full buffer queue to the empty buffer queue.

[0086] FIG. **8** shows a block diagram **800** of a DMS manager **820** that supports backup management of non-relational databases in accordance with aspects of the present disclosure. The DMS manager **820** may be an example of aspects of a DMS manager or a DMS manager **720**, or both, as described herein. The DMS manager **820**, or various components thereof, may be an

example of means for performing various aspects of backup management of non-relational databases as described herein. For example, the DMS manager **820** may include a snapshot scheduling manager **825**, a backup process iteration manager **830**, a synchronization indicator manager **835**, an extractor and mover job initiation manager **840**, a change log file manager **855**, a buffer writing manager **860**, a synchronization document insertion manager **865**, a host API manager **870**, an empty buffer retrieval manager **875**, a full buffer transition manager **880**, a data mover manager **885**, an empty buffer transition manager **890**, or any combination thereof. Each of these components, or components of subcomponents thereof (e.g., one or more processors, one or more memories), may communicate, directly or indirectly, with one another (e.g., via one or more buses, communications links, communications interfaces, or any combination thereof).

[0087] The DMS manager **820** may support data management in accordance with examples as disclosed herein. The snapshot scheduling manager **825** may be configured as or otherwise support a means for obtaining, by a DMS configured to manage backup operations for a non-relational database, an indication to capture a snapshot of the non-relational database from a host of the non-relational database. The backup process iteration manager **830** may be configured as or otherwise support a means for causing, by the DMS, performance by an agent of the DMS at the host of one or more iterations of a backup process until the snapshot is captured. In some examples, to perform an iteration of the backup process, the empty buffer retrieval manager **875** may be configured as or otherwise support a means for retrieving, based on a presence of uncaptured data in the non-relational database, a buffer at the agent from an empty buffer queue for the agent, the buffer writing manager **860** may be configured as or otherwise support a means for writing, based on the buffer being retrieved, data from the non-relational database into the buffer until the buffer is full, where the data includes at least a portion of the uncaptured data, the full buffer transition manager **880** may be configured as or otherwise support a means for transitioning, based at least in part upon filling of the buffer with the data, the buffer to a full buffer queue for the agent, the data mover manager **885** may be configured as or otherwise support a means for moving, based on the buffer being transitioned to the full buffer queue, the data from the buffer to a remote storage environment accessible to the DMS, and the empty buffer transition manager **890** may be configured as or otherwise support a means for transitioning, based on completion of moving the data to the remote storage environment, the buffer from the full buffer queue to the empty buffer queue.

[0088] In some examples, the snapshot is associated with a set of collections of data that are stored at both the host and a second host of the non-relational database, and the backup process iteration manager **830** may be configured as or otherwise support a means for causing, by the DMS, performance of one or more second iterations of the backup process by a second agent of the DMS at the second host until a second portion of the snapshot is captured from the second host, where performance by the agent of the DMS of the one or more iterations of the backup process until the snapshot is captured includes performance of the one or more iterations until a first portion of the snapshot is captured from the host, the first portion including a first subset of collections of the set of collections, and the second portion including a second subset of collections of the set of collections.

[0089] In some examples, the one or more iterations and the one or more second iterations are performed in parallel.

[0090] In some examples, the set of collections of data is also stored at a third host of the non-relational database, and the backup process iteration manager **830** may be configured as or otherwise support a means for causing, by the DMS, performance of one or more third iterations of the backup process by a third agent of the DMS at the third host until a third portion of the snapshot is captured from the second host, where the third portion includes a third subset of collections of the set of collections.

[0091] In some examples, the host is a primary host of the non-relational database. In some examples, the second host and the third host are secondary hosts of the non-relational database that

are configured to store respective copies of the non-relational database from the host.

[0092] In some examples, the synchronization indicator manager **835** may be configured as or otherwise support a means for determining, by the DMS, a presence of a synchronization indicator document in a collection at the host, where causing performance of the one or more iterations is based on determination of the presence of the synchronization indicator document.

[0093] In some examples, the synchronization document insertion manager **865** may be configured as or otherwise support a means for causing insertion, by the DMS and prior to the determination of the presence of the synchronization indicator document in the collection at the host, of the synchronization indicator document into the collection at a second host of the non-relational database, where the second host and the host both store respective copies of the non-relational database.

[0094] In some examples, the synchronization indicator manager **835** may be configured as or otherwise support a means for determining whether a timestamp indicated by the synchronization indicator document is later than a threshold time, where causing performance of the one or more iterations is based on a determination that the timestamp is later than the threshold time.

[0095] In some examples, the extractor and mover job initiation manager **840** may be configured as or otherwise support a means for causing initiation, by the DMS and based on the indication to capture the snapshot, of an extractor job and a mover job for a first collection from a set of collections of data associated with the snapshot. The extractor job may perform, for the first collection and as part of the one or more iterations of the backup process: the retrieving the buffer from the empty buffer queue; the writing the data from the non-relational database into the buffer; and the transitioning the buffer to the full buffer queue. The mover job may perform, for the first collection and as part of the one or more iterations of the backup process: the moving the data from the buffer to the remote storage environment; and the transitioning the buffer from the full buffer queue to the empty buffer queue.

[0096] In some examples, the extractor and mover job initiation manager **840** may be configured as or otherwise support a means for causing initiation, by the DMS and based on the indication to capture the snapshot, of a second extractor job and a second mover job for a second collection from the set of collections. In some examples, the backup process iteration manager **830** may be configured as or otherwise support a means for causing, by the DMS, performance by the agent of the DMS one or more second iterations of the backup process. In some examples, to a second iteration of the one or more second iterations, the empty buffer retrieval manager **875** may be configured as or otherwise support a means for retrieving, by the second extractor job and based on a presence of second uncaptured data in the second collection, a second buffer at the agent from a second empty buffer queue for the agent, the buffer writing manager **860** may be configured as or otherwise support a means for writing, by the second extractor job and based on the second buffer being retrieved, second data from the second collection into the second buffer until the second buffer is full, where the second data includes at least a second portion of the second uncaptured data, the full buffer transition manager **880** may be configured as or otherwise support a means for transitioning, by the second extractor job and based at least in part upon filling of the second buffer with the second data, the second buffer to a second full buffer queue for the agent, the data mover manager **885** may be configured as or otherwise support a means for moving, by the second mover job and based on the second buffer being transitioned to the second full buffer queue, the second data from the second buffer to the remote storage environment, and the empty buffer transition manager **890** may be configured as or otherwise support a means for transitioning, by the second mover job and based on completion of moving the second data to the remote storage environment, the second buffer from the second full buffer queue to the second empty buffer queue.

[0097] In some examples, selecting the buffer from the empty buffer queue by the extractor job is based on the presence of the uncaptured data in the first collection.

[0098] In some examples, the one or more iterations of the backup process for the first collection

and the one or more second iterations of the backup process for the second collection are performed in parallel.

[0099] In some examples, the change log file manager **855** may be configured as or otherwise support a means for updating, by the DMS, the snapshot based on one or more change log files that indicate one or more changes to the non-relational database during a time period in which the one or more iterations are performed.

[0100] In some examples, the host API manager **870** may be configured as or otherwise support a means for performing, by the DMS, an application programming interface call to the host. In some examples, the change log file manager **855** may be configured as or otherwise support a means for receiving, by the DMS from the host in response to the application programming interface call, an indication that the host is configured to capture change log files for the non-relational database, where causing performance of the one or more iterations is based on reception of the indication that the host is configured to capture the change log files.

[0101] In some examples, to support writing data from the non-relational database into the buffer, the buffer writing manager **860** may be configured as or otherwise support a means for writing one or more documents into the buffer.

[0102] In some examples, to support obtaining the indication to capture the snapshot, the snapshot scheduling manager **825** may be configured as or otherwise support a means for determining, by the DMS, that a time period since a last snapshot of the non-relational database satisfies a threshold time period.

[0103] In some examples, to support obtaining the indication to capture the snapshot, the snapshot scheduling manager **825** may be configured as or otherwise support a means for receiving the indication to capture the snapshot from a computing device associated with an administrative account associated with the non-relational database.

[0104] FIG. **9** shows a block diagram **900** of a system **905** that supports backup management of non-relational databases in accordance with aspects of the present disclosure. The system **905** may be an example of or include components of a system **705** as described herein. The system **905** may include components for data management, including components such as a DMS manager **920**, an input information **910**, an output information **915**, a network interface **925**, at least one memory **930**, at least one processor **935**, and a storage **940**. These components may be in electronic communication or otherwise coupled with each other (e.g., operatively, communicatively, functionally, electronically, electrically; via one or more buses, communications links, communications interfaces, or any combination thereof). Additionally, the components of the system **905** may include corresponding physical components or may be implemented as corresponding virtual components (e.g., components of one or more virtual machines). In some examples, the system **905** may be an example of aspects of one or more components described with reference to FIG. **1**, such as a DMS **110**.

[0105] The network interface **925** may enable the system **905** to exchange information (e.g., input information **910**, output information **915**, or both) with other systems or devices (not shown). For example, the network interface **925** may enable the system **905** to connect to a network (e.g., a network **120** as described herein). The network interface **925** may include one or more wireless network interfaces, one or more wired network interfaces, or any combination thereof. In some examples, the network interface **925** may be an example of aspects of one or more components described with reference to FIG. **1**, such as one or more network interfaces **165**.

[0106] Memory **930** may include RAM, ROM, or both. The memory **930** may store computer-readable, computer-executable software including instructions that, when executed, cause the processor **935** to perform various functions described herein. In some cases, the memory **930** may contain, among other things, a basic input/output system (BIOS), which may control basic hardware or software operation such as the interaction with peripheral components or devices. In some cases, the memory **930** may be an example of aspects of one or more components described

with reference to FIG. 1, such as one or more memories **175**.

[0107] The processor **935** may include an intelligent hardware device, (e.g., a general-purpose processor, a DSP, a CPU, a microcontroller, an ASIC, a field programmable gate array (FPGA), a programmable logic device, a discrete gate or transistor logic component, a discrete hardware component, or any combination thereof). The processor **935** may be configured to execute computer-readable instructions stored in a memory **930** to perform various functions (e.g., functions or tasks supporting backup management of non-relational databases). Though a single processor **935** is depicted in the example of FIG. 9, it is to be understood that the system **905** may include any quantity of one or more of processors **935** and that a group of processors **935** may collectively perform one or more functions ascribed herein to a processor, such as the processor **935**. In some cases, the processor **935** may be an example of aspects of one or more components described with reference to FIG. 1, such as one or more processors **170**.

[0108] Storage **940** may be configured to store data that is generated, processed, stored, or otherwise used by the system **905**. In some cases, the storage **940** may include one or more HDDs, one or more SSDs, or both. In some examples, the storage **940** may be an example of a single database, a distributed database, multiple distributed databases, a data store, a data lake, or an emergency backup database. In some examples, the storage **940** may be an example of one or more components described with reference to FIG. 1, such as one or more network disks **180**.

[0109] The DMS manager **920** may support data management in accordance with examples as disclosed herein. For example, the DMS manager **920** may be configured as or otherwise support a means for obtaining, by a DMS configured to manage backup operations for a non-relational database, an indication to capture a snapshot of the non-relational database from a host of the non-relational database. The DMS manager **920** may be configured as or otherwise support a means for causing, by the DMS, performance by an agent of the DMS at the host of one or more iterations of a backup process until the snapshot is captured. In some examples, to perform an iteration of the backup process, the DMS manager **920** may be configured as or otherwise support a means for retrieving, based on a presence of uncaptured data in the non-relational database, a buffer at the agent from an empty buffer queue for the agent, writing, based on the buffer being retrieved, data from the non-relational database into the buffer until the buffer is full, where the data includes at least a portion of the uncaptured data, transitioning, based at least in part upon filling of the buffer with the data, the buffer to a full buffer queue for the agent, moving, based on the buffer being transitioned to the full buffer queue, the data from the buffer to a remote storage environment accessible to the DMS, and transitioning, based on completion of moving the data to the remote storage environment, the buffer from the full buffer queue to the empty buffer queue.

[0110] By including or configuring the DMS manager **920** in accordance with examples as described herein, the system **905** may support techniques for backup management of non-relational databases, which may provide one or more benefits such as, for example, reduced latency, improved user experience, more efficient utilization of computing resources, network resources or both, or improved scalability, among other possibilities.

[0111] FIG. 10 shows a flowchart illustrating a method **1000** that supports backup management of non-relational databases in accordance with aspects of the present disclosure. The operations of the method **1000** may be implemented by a DMS or its components as described herein. For example, the operations of the method **1000** may be performed by a DMS as described with reference to FIGS. 1 through 9. In some examples, a DMS may execute a set of instructions to control the functional elements of the DMS to perform the described functions. Additionally, or alternatively, the DMS may perform aspects of the described functions using special-purpose hardware.

[0112] At **1005**, the method may include obtaining, by a DMS configured to manage backup operations for a non-relational database, an indication to capture a snapshot of the non-relational database from a host of the non-relational database. The operations of **1005** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1005**

may be performed by a snapshot scheduling manager **825** as described with reference to FIG. **8**.

[0113] At **1010**, the method may include causing, by the DMS, performance by an agent of the DMS at the host of one or more iterations of a backup process until the snapshot is captured. In some examples, an iteration of the backup process may include retrieving, based on a presence of uncaptured data in the non-relational database, a buffer at the agent from an empty buffer queue for the agent, writing, based on the buffer being retrieved, data from the non-relational database into the buffer until the buffer is full, where the data includes at least a portion of the uncaptured data, transitioning, based at least in part upon filling of the buffer with the data, the buffer to a full buffer queue for the agent, moving, based on the buffer being transitioned to the full buffer queue, the data from the buffer to a remote storage environment accessible to the DMS, and transitioning, based on completion of moving the data to the remote storage environment, the buffer from the full buffer queue to the empty buffer queue. The operations of **1010** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1010** may be performed by a backup process iteration manager **830** as described with reference to FIG. **8**.

[0114] FIG. **11** shows a flowchart illustrating a method **1100** that supports backup management of non-relational databases in accordance with aspects of the present disclosure. The operations of the method **1100** may be implemented by a DMS or its components as described herein. For example, the operations of the method **1100** may be performed by a DMS as described with reference to FIGS. **1** through **9**. In some examples, a DMS may execute a set of instructions to control the functional elements of the DMS to perform the described functions. Additionally, or alternatively, the DMS may perform aspects of the described functions using special-purpose hardware.

[0115] At **1105**, the method may include obtaining, by a DMS configured to manage backup operations for a non-relational database, an indication to capture a snapshot of the non-relational database from a host of the non-relational database. The operations of **1105** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1105** may be performed by a snapshot scheduling manager **825** as described with reference to FIG. **8**.

[0116] At **1110**, the method may include causing, by the DMS, performance by an agent of the DMS at the host of one or more iterations of a backup process until the snapshot is captured. In some examples, an iteration of the backup process may include retrieving, based on a presence of uncaptured data in the non-relational database, a buffer at the agent from an empty buffer queue for the agent, writing, based on the buffer being retrieved, data from the non-relational database into the buffer until the buffer is full, where the data includes at least a portion of the uncaptured data, transitioning, based at least in part upon filling of the buffer with the data, the buffer to a full buffer queue for the agent, moving, based on the buffer being transitioned to the full buffer queue, the data from the buffer to a remote storage environment accessible to the DMS, and transitioning, based on completion of moving the data to the remote storage environment, the buffer from the full buffer queue to the empty buffer queue. The operations of **1110** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1110** may be performed by a backup process iteration manager **830** as described with reference to FIG. **8**.

[0117] At **1115**, the snapshot is associated with a set of collections of data that are stored at both the host and a second host of the non-relational database, and the method may include causing, by the DMS, performance of one or more second iterations of the backup process by a second agent of the DMS at the second host until a second portion of the snapshot is captured from the second host, where performance by the agent of the DMS of the one or more iterations of the backup process until the snapshot is captured includes performance of the one or more iterations until a first portion of the snapshot is captured from the host, the first portion including a first subset of collections of the set of collections, and the second portion including a second subset of collections of the set of collections. The operations of **1115** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1115** may be performed by a backup process iteration manager **830** as described with reference to FIG. **8**.

[0118] FIG. 12 shows a flowchart illustrating a method 1200 that supports backup management of non-relational databases in accordance with aspects of the present disclosure. The operations of the method 1200 may be implemented by a DMS or its components as described herein. For example, the operations of the method 1200 may be performed by a DMS as described with reference to FIGS. 1 through 9. In some examples, a DMS may execute a set of instructions to control the functional elements of the DMS to perform the described functions. Additionally, or alternatively, the DMS may perform aspects of the described functions using special-purpose hardware.

[0119] At 1205, the method may include obtaining, by a DMS configured to manage backup operations for a non-relational database, an indication to capture a snapshot of the non-relational database from a host of the non-relational database. The operations of 1205 may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of 1205 may be performed by a snapshot scheduling manager 825 as described with reference to FIG. 8.

[0120] At 1210, the method may include determining, by the DMS, a presence of a synchronization indicator document in a collection at the host. The operations of 1210 may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of 1210 may be performed by a synchronization indicator manager 835 as described with reference to FIG. 8.

[0121] At 1215, the method may include causing, by the DMS, performance by an agent of the DMS at the host of one or more iterations of a backup process until the snapshot is captured, where causing performance of the one or more iterations is based on determination of the presence of the synchronization indicator document. In some examples, an iteration of the backup process may include retrieving, based on a presence of uncaptured data in the non-relational database, a buffer at the agent from an empty buffer queue for the agent, writing, based on the buffer being retrieved, data from the non-relational database into the buffer until the buffer is full, where the data includes at least a portion of the uncaptured data, transitioning, based at least in part upon filling of the buffer with the data, the buffer to a full buffer queue for the agent, moving, based on the buffer being transitioned to the full buffer queue, the data from the buffer to a remote storage environment accessible to the DMS, and transitioning, based on completion of moving the data to the remote storage environment, the buffer from the full buffer queue to the empty buffer queue. The operations of 1215 may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of 1215 may be performed by a backup process iteration manager 830 as described with reference to FIG. 8.

[0122] A method for data management by an apparatus is described. The method may include obtaining, by a DMS configured to manage backup operations for a non-relational database, an indication to capture a snapshot of the non-relational database from a host of the non-relational database and causing, by the DMS, performance by an agent of the DMS at the host of one or more iterations of a backup process until the snapshot is captured, where an iteration of the backup process includes retrieving, based on a presence of uncaptured data in the non-relational database, a buffer at the agent from an empty buffer queue for the agent, writing, based on the buffer being retrieved, data from the non-relational database into the buffer until the buffer is full, where the data includes at least a portion of the uncaptured data, transitioning, based at least in part upon filling of the buffer with the data, the buffer to a full buffer queue for the agent, moving, based on the buffer being transitioned to the full buffer queue, the data from the buffer to a remote storage environment accessible to the DMS, and transitioning, based on completion of moving the data to the remote storage environment, the buffer from the full buffer queue to the empty buffer queue.

[0123] An apparatus for data management is described. The apparatus may include one or more memories storing processor executable code, and one or more processors coupled with the one or more memories. The one or more processors may individually or collectively be operable to execute the code to cause the apparatus to obtain, by a DMS configured to manage backup operations for a non-relational database, an indication to capture a snapshot of the non-relational

database from a host of the non-relational database and cause, by the DMS, performance by an agent of the DMS at the host of one or more iterations of a backup process until the snapshot is captured, where, to perform an iteration of the backup process, the one or more processors are individually or collectively operable to execute the code to cause the apparatus to retrieve, based on a presence of uncaptured data in the non-relational database, a buffer at the agent from an empty buffer queue for the agent, write, based on the buffer being retrieved, data from the non-relational database into the buffer until the buffer is full, where the data includes at least a portion of the uncaptured data, transition, based at least in part upon filling of the buffer with the data, the buffer to a full buffer queue for the agent, move, based on the buffer being transitioned to the full buffer queue, the data from the buffer to a remote storage environment accessible to the DMS, and transition, based on completion of moving the data to the remote storage environment, the buffer from the full buffer queue to the empty buffer queue.

[0124] Another apparatus for data management is described. The apparatus may include means for obtaining, by a DMS configured to manage backup operations for a non-relational database, an indication to capture a snapshot of the non-relational database from a host of the non-relational database and means for causing, by the DMS, performance by an agent of the DMS at the host of one or more iterations of a backup process until the snapshot is captured, where the means for an iteration of the backup process include means for retrieving, based on a presence of uncaptured data in the non-relational database, a buffer at the agent from an empty buffer queue for the agent, means for writing, based on the buffer being retrieved, data from the non-relational database into the buffer until the buffer is full, where the data includes at least a portion of the uncaptured data, means for transitioning, based at least in part upon filling of the buffer with the data, the buffer to a full buffer queue for the agent, means for moving, based on the buffer being transitioned to the full buffer queue, the data from the buffer to a remote storage environment accessible to the DMS, and means for transitioning, based on completion of moving the data to the remote storage environment, the buffer from the full buffer queue to the empty buffer queue.

[0125] A non-transitory computer-readable medium storing code for data management is described. The code may include instructions executable by one or more processors to obtain, by a DMS configured to manage backup operations for a non-relational database, an indication to capture a snapshot of the non-relational database from a host of the non-relational database and cause, by the DMS, performance by an agent of the DMS at the host of one or more iterations of a backup process until the snapshot is captured, where the instructions to perform an iteration of the backup process are executable to retrieve, based on a presence of uncaptured data in the non-relational database, a buffer at the agent from an empty buffer queue for the agent, write, based on the buffer being retrieved, data from the non-relational database into the buffer until the buffer is full, where the data includes at least a portion of the uncaptured data, transition, based at least in part upon filling of the buffer with the data, the buffer to a full buffer queue for the agent, move, based on the buffer being transitioned to the full buffer queue, the data from the buffer to a remote storage environment accessible to the DMS, and transition, based on completion of moving the data to the remote storage environment, the buffer from the full buffer queue to the empty buffer queue.

[0126] In some examples of the method, apparatus, and non-transitory computer-readable medium described herein, the snapshot may be associated with a set of collections of data that may be stored at both the host and a second host of the non-relational database and the method, apparatuses, and non-transitory computer-readable medium may include further operations, features, means, or instructions for causing, by the DMS, performance of one or more second iterations of the backup process by a second agent of the DMS at the second host until a second portion of the snapshot may be captured from the second host, where performance by the agent of the DMS of the one or more iterations of the backup process until the snapshot may be captured includes performance of the one or more iterations until a first portion of the snapshot may be captured from the host, the first portion including a first subset of collections of the set of collections, and the second portion

including a second subset of collections of the set of collections.

[0127] In some examples of the method, apparatus, and non-transitory computer-readable medium described herein, the one or more iterations and the one or more second iterations may be performed in parallel.

[0128] In some examples of the method, apparatus, and non-transitory computer-readable medium described herein, the set of collections of data may be also stored at a third host of the non-relational database and the method, apparatuses, and non-transitory computer-readable medium may include further operations, features, means, or instructions for causing, by the DMS, performance of one or more third iterations of the backup process by a third agent of the DMS at the third host until a third portion of the snapshot may be captured from the second host, where the third portion includes a third subset of collections of the set of collections.

[0129] In some examples of the method, apparatus, and non-transitory computer-readable medium described herein, the host may be a primary host of the non-relational database and the second host and the third host may be secondary hosts of the non-relational database that may be configured to store respective copies of the non-relational database from the host.

[0130] Some examples of the method, apparatus, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for determining, by the DMS, a presence of a synchronization indicator document in a collection at the host, where causing performance of the one or more iterations may be based on determination of the presence of the synchronization indicator document.

[0131] Some examples of the method, apparatus, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for causing insertion, by the DMS and prior to the determination of the presence of the synchronization indicator document in the collection at the host, of the synchronization indicator document into the collection at a second host of the non-relational database, where the second host and the host both store respective copies of the non-relational database.

[0132] Some examples of the method, apparatus, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for determining whether a timestamp indicated by the synchronization indicator document may be later than a threshold time, where causing performance of the one or more iterations may be based on a determination that the timestamp may be later than the threshold time.

[0133] Some examples of the method, apparatus, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for causing initiation, by the DMS and based on the indication to capture the snapshot, of an extractor job and a mover job for a first collection from a set of collections of data associated with the snapshot. The extractor job may perform, for the first collection and as part of the one or more iterations of the backup process: the retrieving the buffer from the empty buffer queue; the writing the data from the non-relational database into the buffer; and the transitioning the buffer to the full buffer queue. The mover job may perform, for the first collection and as part of the one or more iterations of the backup process: the moving the data from the buffer to the remote storage environment; and the transitioning the buffer from the full buffer queue to the empty buffer queue.

[0134] Some examples of the method, apparatus, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for causing initiation, by the DMS and based on the indication to capture the snapshot, of a second extractor job and a second mover job for a second collection from the set of collections and causing, by the DMS, performance by the agent of the DMS one or more second iterations of the backup process, where a second iteration of the one or more second iterations includes: retrieving, by the second extractor job and based a presence of second uncaptured data in the second collection, a second buffer at the agent from a second empty buffer queue for the agent; writing, by the second extractor job and based on the second buffer being retrieved, second data from the second collection into the

second buffer until the second buffer is full, where the second data comprises at least a second portion of the second uncaptured data; transitioning, by the second extractor job and based upon filling of the second buffer with the second data, the second buffer to a second full buffer queue for the agent; moving, by the second mover job and based on the second buffer being transitioned to the second full buffer queue, the second data from the second buffer to the remote storage environment; and transitioning, by the second mover job and based on completion of moving the second data to the remote storage environment, the second buffer from the second full buffer queue to the second empty buffer queue.

[0135] Some examples of the method, apparatus, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for selecting the buffer from the empty buffer queue by the extractor job may be based on the presence of the uncaptured data in the first collection.

[0136] In some examples of the method, apparatus, and non-transitory computer-readable medium described herein, the one or more iterations of the backup process for the first collection and the one or more second iterations of the backup process for the second collection may be performed in parallel.

[0137] Some examples of the method, apparatus, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for updating, by the DMS, the snapshot based on one or more change log files that indicate one or more changes to the non-relational database during a time period in which the one or more iterations may be performed.

[0138] Some examples of the method, apparatus, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for performing, by the DMS, an application programming interface call to the host and receiving, by the DMS from the host in response to the application programming interface call, an indication that the host may be configured to capture change log files for the non-relational database, where causing performance of the one or more iterations may be based on reception of the indication that the host may be configured to capture the change log files.

[0139] In some examples of the method, apparatus, and non-transitory computer-readable medium described herein, writing data from the non-relational database into the buffer may include operations, features, means, or instructions for writing one or more documents into the buffer.

[0140] In some examples of the method, apparatus, and non-transitory computer-readable medium described herein, obtaining the indication to capture the snapshot may include operations, features, means, or instructions for determining, by the DMS, that a time period since a last snapshot of the non-relational database satisfies a threshold time period.

[0141] In some examples of the method, apparatus, and non-transitory computer-readable medium described herein, obtaining the indication to capture the snapshot may include operations, features, means, or instructions for receiving the indication to capture the snapshot from a computing device associated with an administrative account associated with the non-relational database.

[0142] It should be noted that the methods described above describe possible implementations, and that the operations and the steps may be rearranged or otherwise modified and that other implementations are possible. Furthermore, aspects from two or more of the methods may be combined.

[0143] The description set forth herein, in connection with the appended drawings, describes example configurations and does not represent all the examples that may be implemented or that are within the scope of the claims. The term “exemplary” used herein means “serving as an example, instance, or illustration,” and not “preferred” or “advantageous over other examples.” The detailed description includes specific details for the purpose of providing an understanding of the described techniques. These techniques, however, may be practiced without these specific details. In some instances, well-known structures and devices are shown in block diagram form in order to

avoid obscuring the concepts of the described examples.

[0144] In the appended figures, similar components or features may have the same reference label. Further, various components of the same type may be distinguished by following the reference label by a dash and a second label that distinguishes among the similar components. If just the first reference label is used in the specification, the description is applicable to any one of the similar components having the same first reference label irrespective of the second reference label.

[0145] Information and signals described herein may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the above description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof.

[0146] The various illustrative blocks and modules described in connection with the disclosure herein may be implemented or performed with a general-purpose processor, a DSP, an ASIC, an FPGA or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor may be a microprocessor, but in the alternative, the processor may be any conventional processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices (e.g., a combination of a DSP and a microprocessor, multiple microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration).

[0147] The functions described herein may be implemented in hardware, software executed by a processor, firmware, or any combination thereof. If implemented in software executed by a processor, the functions may be stored on or transmitted over as one or more instructions or code on a computer-readable medium. Other examples and implementations are within the scope of the disclosure and appended claims. For example, due to the nature of software, functions described above can be implemented using software executed by a processor, hardware, firmware, hardwiring, or combinations of any of these. Features implementing functions may also be physically located at various positions, including being distributed such that portions of functions are implemented at different physical locations. Further, a system as used herein may be a collection of devices, a single device, or aspects within a single device.

[0148] Computer-readable media includes both non-transitory computer storage media and communication media including any medium that facilitates transfer of a computer program from one place to another. A non-transitory storage medium may be any available medium that can be accessed by a general purpose or special purpose computer. By way of example, and not limitation, non-transitory computer-readable media can comprise RAM, ROM, EEPROM, compact disk (CD) ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other non-transitory medium that can be used to carry or store desired program code means in the form of instructions or data structures and that can be accessed by a general-purpose or special-purpose computer, or a general-purpose or special-purpose processor. Also, any connection is properly termed a computer-readable medium. For example, if the software is transmitted from a website, server, or other remote source using a coaxial cable, fiber optic cable, twisted pair, digital subscriber line (DSL), or wireless technologies such as infrared, radio, and microwave, then the coaxial cable, fiber optic cable, twisted pair, DSL, or wireless technologies such as infrared, radio, and microwave are included in the definition of medium. Disk and disc, as used herein, include CD, laser disc, optical disc, digital versatile disc (DVD), floppy disk and Blu-ray disc where disks usually reproduce data magnetically, while discs reproduce data optically with lasers.

Combinations of the above are also included within the scope of computer-readable media.

[0149] As used herein, including in the claims, the article “a” before a noun is open-ended and understood to refer to “at least one” of those nouns or “one or more” of those nouns. Thus, the terms “a,” “at least one,” “one or more,” and “at least one of one or more” may be interchangeable.

For example, if a claim recites “a component” that performs one or more functions, each of the individual functions may be performed by a single component or by any combination of multiple components. Thus, “a component” having characteristics or performing functions may refer to “at least one of one or more components” having a particular characteristic or performing a particular function. Subsequent reference to a component introduced with the article “a” using the terms “the” or “said” refers to any or all of the one or more components. For example, a component introduced with the article “a” shall be understood to mean “one or more components,” and referring to “the component” subsequently in the claims shall be understood to be equivalent to referring to “at least one of the one or more components.”

[0150] Also, as used herein, including in the claims, “or” as used in a list of items (for example, a list of items prefaced by a phrase such as “at least one of” or “one or more of”) indicates an inclusive list such that, for example, a list of at least one of A, B, or C means A or B or C or AB or AC or BC or ABC (i.e., A and B and C). Also, as used herein, the phrase “based on” shall not be construed as a reference to a closed set of conditions. For example, an exemplary step that is described as “based on condition A” may be based on both a condition A and a condition B without departing from the scope of the present disclosure. In other words, as used herein, the phrase “based on” shall be construed in the same manner as the phrase “based at least in part on.”

[0151] The description herein is provided to enable a person skilled in the art to make or use the disclosure. Various modifications to the disclosure will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other variations without departing from the scope of the disclosure. Thus, the disclosure is not limited to the examples and designs described herein but is to be accorded the broadest scope consistent with the principles and novel features disclosed herein.

Claims

1. A method for data management, comprising: obtaining, by a data management system (DMS) configured to manage backup operations for a non-relational database, an indication to capture a snapshot of the non-relational database from a host of the non-relational database; and causing, by the DMS, performance by an agent of the DMS at the host of one or more iterations of a backup process until the snapshot is captured, wherein an iteration of the backup process comprises: retrieving, based at least in part on a presence of uncaptured data in the non-relational database, a buffer at the agent from an empty buffer queue for the agent; writing, based at least in part on the buffer being retrieved, data from the non-relational database into the buffer until the buffer is full, wherein the data comprises at least a portion of the uncaptured data; transitioning, based at least in part upon filling of the buffer with the data, the buffer to a full buffer queue for the agent; moving, based at least in part on the buffer being transitioned to the full buffer queue, the data from the buffer to a remote storage environment accessible to the DMS; and transitioning, based at least in part on completion of moving the data to the remote storage environment, the buffer from the full buffer queue to the empty buffer queue.
2. The method of claim 1, wherein the snapshot is associated with a set of collections of data that are stored at both the host and a second host of the non-relational database, the method further comprising: causing, by the DMS, performance of one or more second iterations of the backup process by a second agent of the DMS at the second host until a second portion of the snapshot is captured from the second host, wherein performance by the agent of the DMS of the one or more iterations of the backup process until the snapshot is captured comprises performance of the one or more iterations until a first portion of the snapshot is captured from the host, the first portion comprising a first subset of collections of the set of collections, and the second portion comprising a second subset of collections of the set of collections.
3. The method of claim 2, wherein the one or more iterations and the one or more second iterations

are performed in parallel.

4. The method of claim 2, wherein the set of collections of data is also stored at a third host of the non-relational database, the method further comprising: causing, by the DMS, performance of one or more third iterations of the backup process by a third agent of the DMS at the third host until a third portion of the snapshot is captured from the second host, wherein the third portion comprises a third subset of collections of the set of collections.

5. The method of claim 4, wherein: the host is a primary host of the non-relational database, and the second host and the third host are secondary hosts of the non-relational database that are configured to store respective copies of the non-relational database from the host.

6. The method of claim 1, further comprising: determining, by the DMS, a presence of a synchronization indicator document in a collection at the host, wherein causing performance of the one or more iterations is based at least in part on determination of the presence of the synchronization indicator document.

7. The method of claim 6, further comprising: causing insertion, by the DMS and prior to the determination of the presence of the synchronization indicator document in the collection at the host, of the synchronization indicator document into the collection at a second host of the non-relational database, wherein the second host and the host both store respective copies of the non-relational database.

8. The method of claim 6, further comprising: determining whether a timestamp indicated by the synchronization indicator document is later than a threshold time, wherein causing performance of the one or more iterations is based at least in part on a determination that the timestamp is later than the threshold time.

9. The method of claim 1, further comprising: causing initiation, by the DMS and based at least in part on the indication to capture the snapshot, of an extractor job and a mover job for a first collection from a set of collections of data associated with the snapshot, wherein the extractor job performs, for the first collection and as part of the one or more iterations of the backup process, the retrieving the buffer from the empty buffer queue, the writing the data from the non-relational database into the buffer, and the transitioning the buffer to the full buffer queue, and wherein the mover job performs, for the first collection and as part of the one or more iterations of the backup process, the moving the data from the buffer to the remote storage environment and the transitioning the buffer from the full buffer queue to the empty buffer queue.

10. The method of claim 9, further comprising: causing initiation, by the DMS and based at least in part on the indication to capture the snapshot, of a second extractor job and a second mover job for a second collection from the set of collections; and causing, by the DMS, performance by the agent of the DMS one or more second iterations of the backup process, wherein a second iteration of the one or more second iterations comprises: retrieving, by the second extractor job and based at least in part on a presence of second uncaptured data in the second collection, a second buffer at the agent from a second empty buffer queue for the agent; writing, by the second extractor job and based at least in part on the second buffer being retrieved, second data from the second collection into the second buffer until the second buffer is full, wherein the second data comprises at least a second portion of the second uncaptured data; transitioning, by the second extractor job and based at least in part upon filling of the second buffer with the second data, the second buffer to a second full buffer queue for the agent; moving, by the second mover job and based at least in part on the second buffer being transitioned to the second full buffer queue, the second data from the second buffer to the remote storage environment; and transitioning, by the second mover job and based at least in part on completion of moving the second data to the remote storage environment, the second buffer from the second full buffer queue to the second empty buffer queue.

11. The method of claim 10, wherein selecting the buffer from the empty buffer queue by the extractor job is based at least in part on the presence of the uncaptured data in the first collection.

12. The method of claim 10, wherein the one or more iterations of the backup process for the first

collection and the one or more second iterations of the backup process for the second collection are performed in parallel.

13. The method of claim 1, further comprising: updating, by the DMS, the snapshot based at least in part on one or more change log files that indicate one or more changes to the non-relational database during a time period in which the one or more iterations are performed.

14. The method of claim 13, further comprising: performing, by the DMS, an application programming interface call to the host; and receiving, by the DMS from the host in response to the application programming interface call, an indication that the host is configured to capture change log files for the non-relational database, wherein causing performance of the one or more iterations is based at least in part on reception of the indication that the host is configured to capture the change log files.

15. The method of claim 1, wherein writing data from the non-relational database into the buffer comprises: writing one or more documents into the buffer.

16. The method of claim 1, wherein obtaining the indication to capture the snapshot comprises: determining, by the DMS, that a time period since a last snapshot of the non-relational database satisfies a threshold time period.

17. The method of claim 1, wherein obtaining the indication to capture the snapshot comprises: receiving the indication to capture the snapshot from a computing device associated with an administrative account associated with the non-relational database.

18. An apparatus for data management, comprising: one or more memories storing processor-executable code; and one or more processors coupled with the one or more memories and individually or collectively operable to execute the code to cause the apparatus to: obtain, by a data management system (DMS) configured to manage backup operations for a non-relational database, an indication to capture a snapshot of the non-relational database from a host of the non-relational database; and cause, by the DMS, performance by an agent of the DMS at the host of one or more iterations of a backup process until the snapshot is captured, wherein, to perform an iteration of the backup process, the one or more processors are individually or collectively operable to execute the code to cause the apparatus to: retrieve, based at least in part on a presence of uncaptured data in the non-relational database, a buffer at the agent from an empty buffer queue for the agent; write, based at least in part on the buffer being retrieved, data from the non-relational database into the buffer until the buffer is full, wherein the data comprises at least a portion of the uncaptured data; transition, based at least in part upon filling of the buffer with the data, the buffer to a full buffer queue for the agent; move, based at least in part on the buffer being transitioned to the full buffer queue, the data from the buffer to a remote storage environment accessible to the DMS; and transition, based at least in part on completion of moving the data to the remote storage environment, the buffer from the full buffer queue to the empty buffer queue.

19. The apparatus of claim 18, wherein the snapshot is associated with a set of collections of data that are stored at both the host and a second host of the non-relational database, and wherein the one or more processors are individually or collectively further operable to execute the code to cause the apparatus to: cause, by the DMS, performance of one or more second iterations of the backup process by a second agent of the DMS at the second host until a second portion of the snapshot is captured from the second host, wherein performance by the agent of the DMS of the one or more iterations of the backup process until the snapshot is captured comprises performance of the one or more iterations until a first portion of the snapshot is captured from the host, the first portion comprising a first subset of collections of the set of collections, and the second portion comprising a second subset of collections of the set of collections.

20. A non-transitory computer-readable medium storing code for data management, the code comprising instructions executable by one or more processors to: obtain, by a data management system (DMS) configured to manage backup operations for a non-relational database, an indication to capture a snapshot of the non-relational database from a host of the non-relational database; and

cause, by the DMS, performance by an agent of the DMS at the host of one or more iterations of a backup process until the snapshot is captured, wherein, to perform an iteration of the backup process, the instructions are executable by the one or more processors to: retrieve, based at least in part on a presence of uncaptured data in the non-relational database, a buffer at the agent from an empty buffer queue for the agent; write, based at least in part on the buffer being retrieved, data from the non-relational database into the buffer until the buffer is full, wherein the data comprises at least a portion of the uncaptured data; transition, based at least in part upon filling of the buffer with the data, the buffer to a full buffer queue for the agent; move, based at least in part on the buffer being transitioned to the full buffer queue, the data from the buffer to a remote storage environment accessible to the DMS; and transition, based at least in part on completion of moving the data to the remote storage environment, the buffer from the full buffer queue to the empty buffer queue.
